

Historical Flight Visualization in Blender

Turn a **real airline flight** into an animated fly-along on a textured 3D Earth

External OpenSky preprocessor → `flight.json` → one-click Blender add-on

The Idea

Take a **real airline flight** and replay it as a smooth fly-along on a 3D Earth.

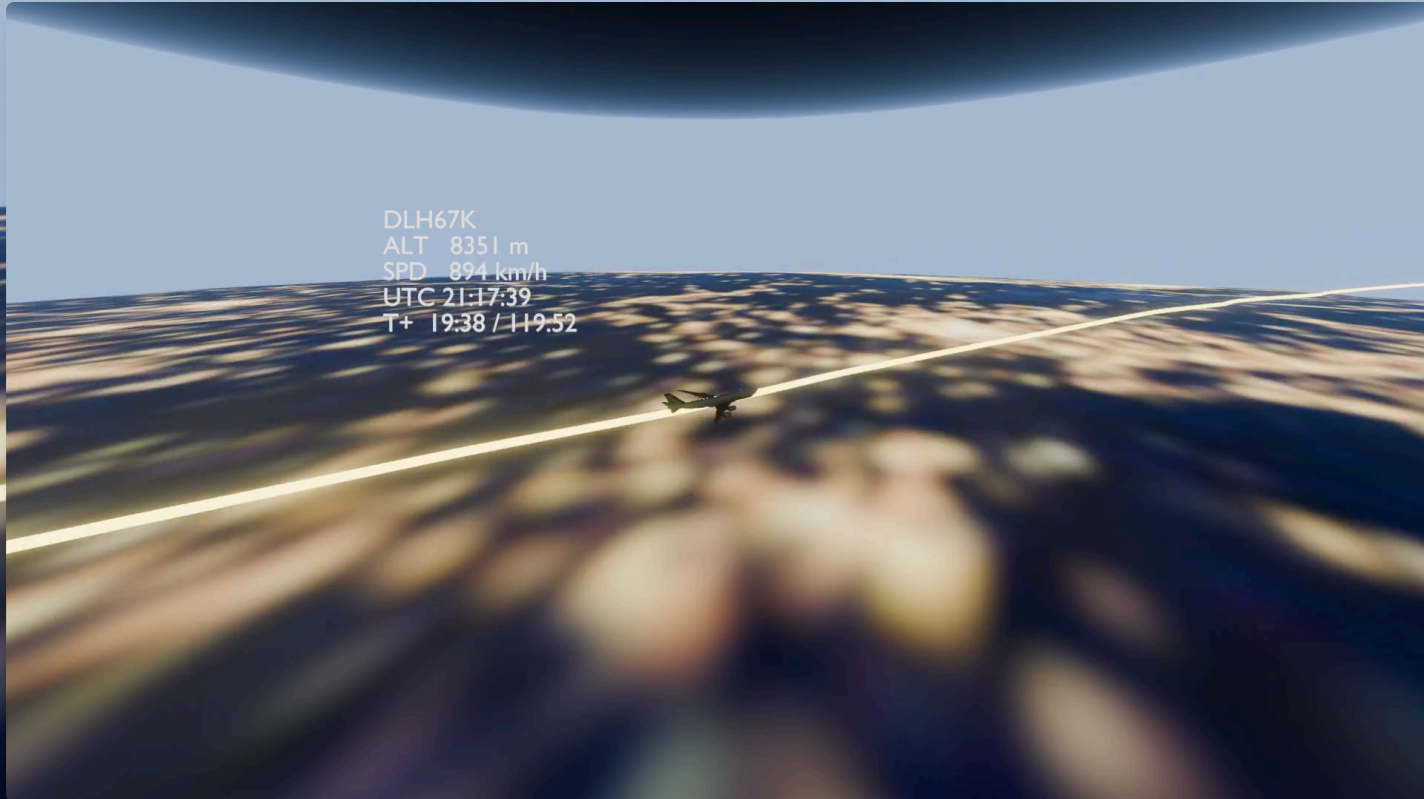
Two parts

- **Outside Blender** – a Python tool downloads the flight from the **OpenSky Network**, cleans it, and writes a small `flight.json`
- **Inside Blender** – a **one-click add-on** reads that file and builds the whole scene

Why two parts?

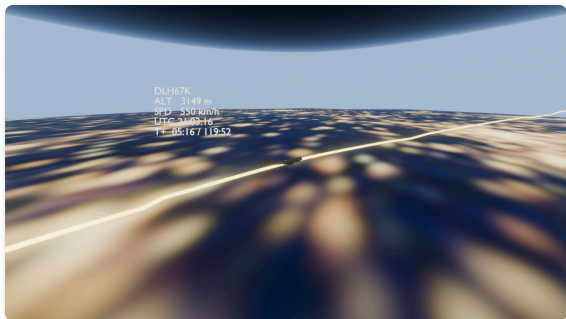
- Network and login code stay out of Blender
- The two parts meet at one simple file:
`flight.json`
- Once downloaded, a flight works **offline**, every time

Final Prototype

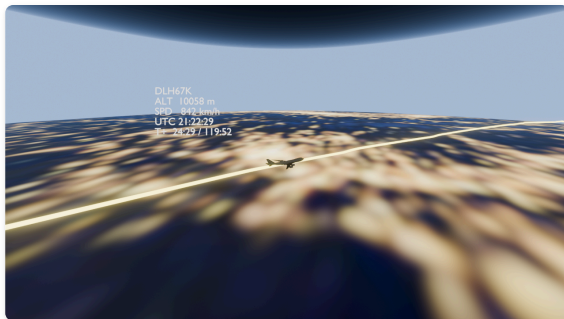


Chase camera – live HUD, route trail, atmosphere glow, night lighting

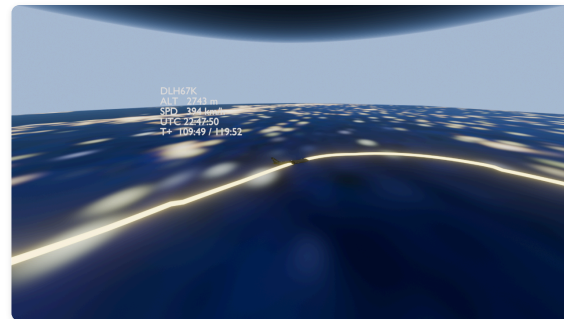
What's in the Scene



Airport at departure



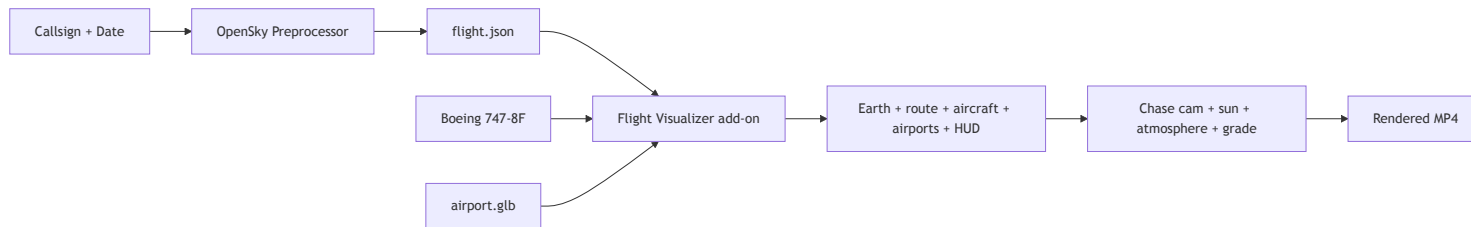
Banks into turns



Descent into arrival

Detailed Earth (day / night / mountains) · airports at both ends · the sun placed by the flight's **real time of day** · the plane pitches and banks · on-screen readout · night light · atmosphere and color polish · **one-click video export**

Architecture



Everything connects through one file: `flight.json`. A small checker makes sure the file is valid before Blender uses it. You can also **fetch a flight straight from the add-on** – it runs the Python tool for you.

Challenge: getting the data from OpenSky

The tricky parts

- You can't search by flight number like "LH401" – the data is keyed to the **aircraft's radio ID** (icao24)
- So I look up the aircraft first, then get its track
- Login uses a **token** (OAuth2)
- Only the **last ~30 days** of flights are available

How I handled it

- Enter a **callsign + departure airport**, or the aircraft ID directly
- If the date is too old, stop early with a **clear message**
- Speed isn't in the data, so I **compute it** from the points
- The download code stays **outside Blender**; the add-on finds the right Python automatically

Cleaning & smoothing the flight data

The raw data is rough: points arrive unevenly, jitter around, and have long gaps. It goes through a small **pipeline** before the plane ever moves:

1. **Place on the globe** – turn each lat/lon/altitude into a 3D point, measured from the **ground right below** it
2. **De-jitter** – average each point with its neighbours to smooth the GPS wobble, but **keep its height** (so it can't sag into the Earth)
3. **Follow the curve** – between points, move along the **curve of the Earth**, not a straight line through it
4. **Even out the timing** – re-sample the path at **regular intervals**, so dense spots and long gaps are treated the same
5. **Equal distance per frame** – finally, space the points so the plane moves the **same distance each frame** → steady speed

Techniques: radius-preserving moving average · spherical (great-circle) interpolation · uniform-time resampling · constant arc-length re-parametrization

Result – no more surging, stalling, or flying underground; the plane moves at a steady, even speed.

Realistic plane rotation

The data only gives **position**, not which way the plane faces. I build the orientation from the path itself, so it flies naturally.

Heading Nose points toward the **next point** on the path – so it always faces where it's going.

Pitch That direction includes going up or down, so the nose **tilts up on climb** and **down on descent** for free.

Bank When the path curves, the plane **leans into the turn** – the sharper the turn, the more it rolls (capped at 30°, like a real airliner).

"Up" always points **away from the Earth's center**, so the plane sits level on the globe instead of drifting as it crosses the world.

Roadmap

Done

- Download flights from OpenSky, save as `flight.json`
- One-click add-on (+ **fetch from the UI**)
- Smooth, steady motion that stays above the ground
- Chase camera (level, never underground)
- Plane pitches and banks in turns
- Airports at both ends
- On-screen readout (callsign, altitude, speed, time)
- Night light · atmosphere · nicer colors
- Save to video · tests · easy install · docs

Future work

- **Weather layer** (clouds, wind) – left out for now
- **Older flights (30+ days)** – needs special OpenSky access
- More camera angles and motion blur
- Sharper Earth textures for close-ups

Thank you

A real airline flight → a smooth, cinematic fly-along on a 3D Earth, built by a **one-click Blender add-on**.

Try it

- `./blender/build_addon.sh` → install the zip
- Open `flight_visualization.blend`
- **N ▶ Flight** → Fetch or Load & Build → **Render Video**

Repo

- Add-on: `blender/flight_viz_addon/`
- Preprocessor + tests: `preprocess/`
- Design notes: `DESIGN.md`

github.com/erinc-emre/DBP

Questions?